

These are the relevant files/folders and data:

Model training

1. **Scalar correction training** was done using this file: `train_pair_correction_new.py`. This created the model weights file: `pair_correction_nn.pt` and training log: `training_log.png`. This is where the issue is raised on softening correction.

Ablations (Results I Section 5.1)

2. The first model was developed before adaptive: The folder with code and results are here: `set 2_before_adaptive`. This is **Run A** in Table 3 of the paper.
3. Then tried the adaptive sub-stepping: The folder with code and results are here: `set 3_after_adaptive`. This folder has [Figure 5 and 6](#) of the paper. This is **Run D** in Table 3 of the paper. Comparing Run A and Run D produced [Figure 1](#) of the paper (created separately)
4. **Run B** in table 3 of paper: Rather than using analytical direction, use the learned vector NN. The files are: `train_vector_nn.py`, `vector_nn.pt`, `training_log_vector.png`. The results are in this folder: `comparison_out` (which have [Figure 2 and 3](#) of the paper).
5. **Run C** in Table 3 of paper: no NN case. File used: `no_nn_ablation.py` and results are in `comparison_out_no_nn` folder. It also has [Figure 4](#) of the paper.

Results II (section 5.2)

6. Multi Initial condition: File used: `multi_ic_eval_v3.py`. Results in: `multi_ic_out_v3`. Table 5 of the paper. [Figures 7 and 8](#).
7. Population modeling (section 5.2.4): in `misc` folder.

Results III

8. Energy conservation: `energy_drift_eval.py` and `energy_drift_scaling.py`. Results in `energy_drift_scaling`. Table 6 and [Figure 9](#).
9. Seed robustness: File: `seed_robustness.py`. Results in `seed_robustness`. Table 7 and [Figure 10](#).

Results IV

10. Speed-Accuracy: These are the results from the same Run D presented for speed-accuracy. The folder with code and results are here: `set 3_after_adaptive`. [Figure 11](#) and Table 8

Appendix A and B

11. In the `misc` folder.